

DEMO PRESENTATION

HubSpot CRM Sandbox Demo

HubSpot CRM Sandbox 示範簡報

A lightweight CRM demo built in C# for HubSpot workflows, designed to show object creation, deal linking, and CSV import/export in a clean interview-ready format.

這是一個以 C# 開發的輕量 CRM 示範系統，用來展示 HubSpot 物件建立、商機關聯，以及 CSV 匯入匯出流程。

C# / ASP.NET Core

HubSpot API

CSV Import / Export

Validation Workflow

4

Core demo panels
核心展示面板

3

Main object types
主要物件類型

1

Live HubSpot sandbox
真實 HubSpot 測試環境

HubSpot CRM Sandbox

Follow the four steps below in order. Create the company and contact first, then create a deal, then link the deal to the records you already created.

LIVE HUBSPOT ACCOUNT

Reads and writes real records in your sandbox portal. Dropdowns are populated from live HubSpot data so you can't submit invalid values.

STEP 1

Create a company

Start with the organization the deal belongs to.

STEP 2

Create a contact

Add the person who talks to sales at that company.

STEP 3

Create a deal

Pick a real pipeline and stage -- no typing blind IDs.

STEP 4

Link the records

Connect deal -- company and deal -- contact.

WORKSPACE

Deals

23

Contacts

3

Companies

1

Links

--

Import / Export

CSV

Deals

Stages come from HubSpot

Create deals with valid pipeline metadata, search by any property, then use "Use in Links" to pre-select a deal in the Links tab.

Create Deal

The Stage dropdown updates when you change the Pipeline.

Deal name required Amount

AI rack upgrade 25000

Pipeline Stage

Sales Pipeline (de Appointment Schv

Close date

DD.MM.YYYY

Create deal Refresh data

Search Deals

Filter the table on the right without leaving the page.

Property Operator

dealstage EQ -- equals

Value

appointmentscheduled

For HAS_PROPERTY / NOT_HAS_PROPERTY leave value empty.

Search Show all

Deal Records

Click "Use in Links" to pre-select a deal.

23 deals

ID	NAME	STAGE	AMOUNT	PIPELINE	ACTION
320981898946	ERP upgrade 15	--	29400	--	Use in Links
321021445835	SaaS licence 4	--	13400	--	Use in Links
321021445837	ERP upgrade 17	--	1700	--	Use in Links
321074908870	Data warehouse 22	--	17600	--	Use in Links
321183072954	SaaS licence 7	--	38900	--	Use in Links
321183072955	CRM rollout 12	--	25600	--	Use in Links
321184826085	AI pilot 10	--	29900	--	Use in Links
321184826089	Cloud migration 20	--	41100	--	Use in Links
321184826990	Data warehouse 21	--	31100	--	Use in Links
321202298608	Data warehouse 6	--	47400	--	Use in Links
321202298610	AI pilot 19	--	3100	--	Use in Links
321225608907	tester	qualifiedtobuy	--	default	Use in Links
321243678409	SaaS licence 8	--	42900	--	Use in Links
321243678410	API integration 14	--	3600	--	Use in Links
321247271667	ERP upgrade 3	--	37200	--	Use in Links
321249431255	SaaS licence 1	--	47600	--	Use in Links
321249431257	CRM rollout 16	--	11000	--	Use in Links
321249469114	SaaS licence 2	--	15000	--	Use in Links
321249469117	SaaS licence 18	--	34100	--	Use in Links
321261631180	SaaS licence 13	--	24100	--	Use in Links
321261683396	Mobile app 5	--	41500	--	Use in Links
321261705945	SaaS licence 11	--	34400	--	Use in Links
321262307915	CRM rollout 9	--	16600	--	Use in Links

API Result OK

Raw JSON response from the last API call.

Refresh everything

Clear

```
{
  "message": "Data refreshed.",
  "counts": {
    "deals": 23
  }
}
```

```
"contacts": 3,  
"companies": 1  
}
```

English: The UI reads and writes to a live HubSpot sandbox account, while keeping the workflow compact for demo use.

中文：此介面會讀寫 HubSpot 測試帳號，但仍保持簡潔，方便做示範與說明。

WORKFLOW

Demo Flow / 示範流程

The demo follows a practical CRM sequence: create the master data first, then create the deal, and finally link all records together.

此示範依照實務 CRM 流程設計：先建立主資料，再建立商機，最後完成關聯。

1

建立公司資料，作為商機所屬組織。

Create the company record as the organization behind the deal.

2

建立聯絡人資料，對應實際窗口或負責人。

Create the contact record for the real person tied to the opportunity.

3

建立商機，設定金額、流程與階段。

Create the deal and set amount, pipeline, and stage.

4

將商機與公司、聯絡人建立關聯，形成完整 **CRM** 關係。

Associate the deal with the company and contact to complete the CRM relationship.

DEAL MANAGEMENT

Deals Panel / 商機管理畫面

The deals screen supports create, search, live list refresh, and result inspection from a single panel.

商機頁面可在同一個面板完成建立、查詢、清單更新與結果檢視。

A Live pipeline validation / 即時流程驗證

Pipeline and stage values come from HubSpot, so invalid stage combinations are blocked before submission.

流程與階段選項直接來自 HubSpot，可在送出前避免無效組合。

B Searchable deal list / 可查詢的商機清單

The user can filter deals by property and inspect the returned records immediately.

使用者可依欄位條件查詢商機，並立即查看回傳結果。

C Result transparency / 結果透明化

The raw API result is shown for demos, debugging, and interview explanation.

系統會顯示原始 API 結果，方便示範、除錯與面試解說。

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Connect deal -- company and deal -- contact.

WORKSPACE

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Links

--

Import / Export

CSV

Deals

Stages come from HubSpot

Create deals with valid pipeline metadata, search by any property, then use "Use in Links" to pre-select a deal in the Links tab.

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The Stage dropdown updates when you change the Pipeline.

Deal name required Amount

AI rack upgrade 25000

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{
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  "counts": {
    "deals": 23
  }
}
```

English: This is the main operational screen for deal creation and search.

中文：這是商機建立與查詢的主要操作畫面。

MASTER DATA

HubSpot CRM Sandbox

Follow the four steps below in order. Create the company and contact first, then create a deal, then link the deal to the records you already created.

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STEP 1

Create a company. This is the first step in the process. It sets up the organization and creates links to the other steps.

STEP 2

Create a contact. This is the second step in the process. It sets up the person who is associated with the company.

STEP 3

Create a deal. This is the third step in the process. It sets up the transaction that is associated with the company and contact.

STEP 4

Link the records. This is the final step in the process. It links the company, contact, and deal together.

Company and Contact Setup / 公司與聯絡人建立

In CRM practice, clean master data is the foundation for deals, reporting, and downstream integrations.

在 CRM 實務中，乾淨的主資料是商機、報表與後續整合的基礎。

1

Controlled input / 受控輸入

Dropdowns reduce invalid entries and reflect the live HubSpot schema.

下拉選單可減少錯誤輸入，並直接反映 HubSpot 的實際欄位規則。

2

Reusable master records / 可重複使用的主資料

Once created, companies and contacts can be reused across multiple deals and links.

公司與聯絡人建立後，可在多筆商機與關聯流程中重複使用。

3

Closer to integration work / 更接近整合實務

This mirrors how integration engineers prepare CRM entities before deal synchronization.

這也符合整合工程師在同步商機前，先整理 CRM 主資料的實務做法。

English: Company creation uses controlled fields such as industry options from HubSpot.

中文：公司建立頁面採用受控欄位，例如 industry 會讀取 HubSpot 的有效選項。

ASSOCIATIONS

Linking Records / 關聯建立

The linking screen demonstrates how a deal becomes meaningful only after it is associated with the correct company and contact.

關聯畫面展示了商機必須與正確的公司與聯絡人綁定後，才真正具備商業意義。

1 Deal-to-company link / 商機對公司關聯

Shows which organization owns or drives the opportunity.

顯示該商機屬於哪一家公司或由哪個組織推動。

2 Deal-to-contact link / 商機對聯絡人關聯

Tracks the person involved in communication, negotiation, or approval.

追蹤參與聯繫、談判或核准的人員。

3 Good demo for CRM logic / 適合展示 CRM 邏輯

This is a useful screen to explain CRM relationships to technical and non-technical viewers.

此畫面很適合向技術與非技術觀眾說明 CRM 關係模型。

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STEP 4

Link the records

Connect deal → company and deal → contact.

WORKSPACE

Deals

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1

Links

0

Import / Export

CSV

Links

No hand-typed IDs

A link never creates a record — it connects an existing deal to an existing contact or company. Every picker is populated from records already in HubSpot.

Link Deal → Contact

Select a real deal and a real contact, then create the link.

Deal

ERP upgrade 15 [320981898946] · no stage

Contact

Maria Johnson (Sample Contact) [472570274533]

Link to contact

Link Deal → Company

Select a real deal and a real company, then create the link.

Deal

ERP upgrade 15 [320981898946] · no stage

Company

Company [318701590222]

Link to company

View Existing Links

Pick a deal, then load its linked contacts or companies.

Deal

ERP upgrade 15 [320981898946] · no stage

Show linked contacts

Show linked companies

Select a deal above and click a button to see its links.

How associations work

In HubSpot, deals don't contain contact or company data — they reference them through separate association records. This panel sends a PUT request to create the link and a GET to read it back.

Association type IDs

HubSpot uses fixed numeric IDs per object pair: deal→contact = 3, deal→company = 5.

Why dropdowns instead of text inputs

Associations fail silently if you reference an ID that doesn't exist. Dropdowns eliminate that class of error entirely.

API Result

OK

Raw JSON response from the last API call.

Refresh everything

Clear

```
{
  "message": "Data refreshed.",
  "counts": {
    "deals": 23,
    "contacts": 3,
    "companies": 1
  }
}
```

English: The links tab binds existing records instead of creating guessed IDs manually.

中文：Links 頁面會讓使用者直接選擇現有資料，而不是手動猜測 ID。

IMPORT / EXPORT

HubSpot CRM Sandbox

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Reads and writes real records in your sandbox portal. Dropdowns are populated from live HubSpot data so you can't submit invalid values.

STEP 1

Create a company
Create a company with the name and GUID.

STEP 2

Create a contact
Create a contact with the name and email.

STEP 3

Create a deal
Create a deal with the name and stage -- no typing blind IDs.

STEP 4

Link the records
Connect deal --> company and deal --> contact.

CSV Workflow / CSV 匯入匯出流程

The demo supports template download, export, preview validation, and apply-to-HubSpot in one guided flow.

這個 demo 支援樣板下載、資料匯出、預覽驗證，以及套用到 HubSpot 的完整流程。

A Preview before write / 先預覽再寫入

The import preview checks rows before any write operation happens.

匯入預覽會先檢查每一列資料，再決定是否進行寫入。

B Round-trip pattern / Round-trip 模式

Users can export records, edit the CSV, preview changes, and then apply them back to HubSpot.

使用者可先匯出資料，再編輯 CSV，之後預覽並回寫到 HubSpot。

C Integration-oriented design / 面向整合的設計

This is the part that best matches CRM/ERP integration interview requirements.

這部分最能對應 CRM/ERP 整合相關職缺的面試需求。

API Result

Raw JSON response from the last API call.

Refresh everything

Clear

```
{
  "message": "Data refreshed.",
  "counts": {
    "deals": 23,
    "contacts": 3,
    "companies": 1
  }
}
```

English: CSV import/export is the most integration-oriented part of the demo.

中文：CSV 匯入匯出是這個 demo 中最接近整合工程實務的一部分。

ARCHITECTURE

Technical Architecture / 技術架構

The current demo is intentionally lightweight, but it already shows a practical separation between UI, API endpoints, validation logic, and HubSpot integration.

目前版本刻意保持輕量，但已具備 UI、API、驗證邏輯與 HubSpot 整合層之間的基本分工。

Browser UI / 瀏覽器介面

Form input, list rendering, import preview, result inspection.



ASP.NET Core Demo API / ASP.NET Core 示範 API

Receives actions from the UI and exposes bootstrap, create, search, associate, export, and import endpoints.



Validation & CSV Service / 驗證與 CSV 服務

Checks headers, required fields, stages, industries, and import readiness before apply.



HubSpot Sandbox Portal / HubSpot 測試帳號

Stores the live demo records and returns the actual CRM response payloads.

1 Current scope / 目前範圍

Live CRM demo, object creation, associations, CSV preview/apply.

真實 CRM 示範、物件建立、關聯、CSV 預覽與套用。

2 Professional next step / 下一步專業化方向

Add SQL staging, import history, row-level errors, and audit tracking.

加入 SQL 暫存層、匯入歷史、逐列錯誤與稽核紀錄。

3 Interview value / 面試價值

The demo already shows CRM thinking, API usage, data validation, and integration flow design.

此 demo 已能展示 CRM 思維、API 使用、資料驗證與整合流程設計。